

Exercise Sheet 10 — Final Answers Only

Each item shows the antiderivative next to the integral.

1. Answers

$$(a) \int (2x + 3e^x - x\sqrt{x}) dx = \boxed{x^2 + 3e^x - \frac{2}{5}x^{5/2} + C}$$

$$(b) \int \left(5 \cos x + 3^x + \frac{1}{2\sqrt[3]{x}} \right) dx = \boxed{5 \sin x + \frac{3^x}{\ln 3} + \frac{3}{4}x^{2/3} + C}$$

$$(c) \int \frac{(18 + 2x^2) \sin x + 5}{x^2 + 9} dx = \boxed{-2 \cos x + \frac{5}{3} \arctan\left(\frac{x}{3}\right) + C}$$

$$(d) \int \frac{(\sqrt{x} + 2)^2}{\sqrt{x}} dx = \boxed{\frac{2}{3}x^{3/2} + 4x + 8\sqrt{x} + C}$$

$$(e) \int \frac{3 - \cos^2 x}{\cos^2 x} dx = \boxed{3 \tan x - x + C}$$

$$(f) \int \sqrt[3]{5 - 6x} dx = \boxed{-\frac{1}{8} (5 - 6x)^{4/3} + C}$$

$$(g) \int \frac{2x^3 + 8x - 5}{x^2 + 4} dx = \boxed{x^2 - \frac{5}{2} \arctan\left(\frac{x}{2}\right) + C}$$

$$(h) \int 3^{2-3x} dx = \boxed{-\frac{3^{1-3x}}{\ln 3} + C}$$

$$(i) \int \left(\frac{9}{9 + x^2} + \frac{5}{x^2} \right) dx = \boxed{3 \arctan\left(\frac{x}{3}\right) - \frac{5}{x} + C}$$

$$(j) \int \frac{dx}{1 + 2x^2} = \boxed{\frac{1}{\sqrt{2}} \arctan(\sqrt{2}x) + C}$$

$$(k) \int \frac{dx}{\sqrt{36 - 3x^2}} = \boxed{\frac{1}{\sqrt{3}} \arcsin\left(\frac{x}{2\sqrt{3}}\right) + C}$$

2. Answers

$$(a) \int e^{-2x+1} dx = \boxed{-\frac{1}{2}e^{-2x+1} + C}$$

$$(b) \int \frac{dx}{1 - 3x} = \boxed{-\frac{1}{3} \ln |1 - 3x| + C}$$

$$(c) \int \frac{\tan^2 x + 3x \sin^2 x}{\sin^2 x} dx = \boxed{\tan x + \frac{3}{2}x^2 + C}$$

$$(d) \int \frac{dx}{9 - 4x^2} = \boxed{\frac{1}{12} \ln \left| \frac{3 + 2x}{3 - 2x} \right| + C}$$

$$(e) \int \frac{1 + x + 3x^2}{\sqrt{x}} dx = \boxed{2\sqrt{x} + \frac{2}{3}x^{3/2} + \frac{6}{5}x^{5/2} + C}$$

$$(f) \int \left(e^{5x} - \frac{2x^2 + 5}{1 + x^2} \right) dx = \boxed{\frac{1}{5}e^{5x} - 2x - 3 \arctan x + C}$$

$$(g) \int \frac{dx}{(5 - 3x)^2} = \boxed{\frac{1}{3(5 - 3x)} + C}$$

$$(h) \int \frac{dx}{3 - 16x^2} = \boxed{\frac{1}{8\sqrt{3}} \ln \left| \frac{\sqrt{3} + 4x}{\sqrt{3} - 4x} \right| + C}$$

$$(i) \int \frac{dx}{\sqrt{4 - 3x^2}} = \boxed{\frac{1}{\sqrt{3}} \arcsin \left(\frac{\sqrt{3}x}{2} \right) + C}$$

$$(j) \int \frac{dx}{\sqrt{1 - (4x + 2)^2}} = \boxed{\frac{1}{4} \arcsin(4x + 2) + C}$$

$$(k) \int \frac{dx}{1 - 5x^2} = \boxed{\frac{1}{2\sqrt{5}} \ln \left| \frac{1 + \sqrt{5}x}{1 - \sqrt{5}x} \right| + C}$$

3. Answers (substitution)

$$(a) \int \frac{4x^3 - 1}{x^4 - x + 5} dx = \boxed{\ln |x^4 - x + 5| + C}$$

$$(b) \int \frac{\sqrt[4]{\ln x}}{x} dx = \boxed{\frac{4}{5}(\ln x)^{5/4} + C}$$

$$(c) \int \frac{3x}{(x^2 + 4)^3} dx = \boxed{-\frac{3}{4(x^2 + 4)^2} + C}$$

$$(d) \int \frac{e^{\tan x}}{\cos^2 x} dx = \boxed{e^{\tan x} + C}$$

$$(e) \int \frac{\cos x}{\sqrt{\sin^2 x + 3}} dx = \boxed{\ln \left| \sin x + \sqrt{\sin^2 x + 3} \right| + C}$$

$$(f) \int \frac{\sqrt{\arctan x}}{1 + x^2} dx = \boxed{\frac{2}{3}(\arctan x)^{3/2} + C}$$

$$(g) \int \sqrt[3]{2 + 7 \sin x} \cos x dx = \boxed{\frac{3}{28}(2 + 7 \sin x)^{4/3} + C}$$

$$(h) \int \frac{1}{x^2} \cos \left(\frac{1}{x} \right) dx = \boxed{-\sin \left(\frac{1}{x} \right) + C}$$

$$(i) \int \frac{\cos(5x)}{\sin^2(5x) + 1} dx = \boxed{\frac{1}{5} \arctan(\sin(5x)) + C}$$

$$(j) \int \frac{\ln x + 2}{2x} dx = \boxed{\frac{(\ln |x|)^2}{4} + \ln |x| + C}$$

$$(k) \int \frac{dx}{\sin^2 x \sqrt{3 - \cot x}} = \boxed{2\sqrt{3 - \cot x} + C}$$

4. Answers (by parts)

$$(a) \int e^{-x}(x^2 + 5) dx = \boxed{-e^{-x}(x^2 + 2x + 7) + C}$$

$$(b) \int \arcsin x dx = \boxed{x \arcsin x + \sqrt{1 - x^2} + C}$$

$$(c) \int x \sin(3x) dx = \boxed{-\frac{x \cos(3x)}{3} + \frac{\sin(3x)}{9} + C}$$

$$(d) \int e^x \sin(3x) dx = \boxed{\frac{e^x}{10}(\sin(3x) - 3 \cos(3x)) + C}$$

$$(e) \int x \cdot 10^x dx = \boxed{\frac{x 10^x}{\ln 10} - \frac{10^x}{(\ln 10)^2} + C}$$

$$(f) \int (x - 2) \ln(2x) dx = \boxed{\left(\frac{x^2}{2} - 2x\right) \ln(2x) - \frac{x^2}{4} + 2x + C}$$

$$(g) \int \arccos(2x) dx = \boxed{x \arccos(2x) - \frac{1}{2}\sqrt{1 - 4x^2} + C}$$

$$(h) \int \ln(x + 2) dx = \boxed{(x + 2) \ln(x + 2) - (x + 2) + C}$$

$$(i) \int \cos(\ln x) dx = \boxed{\frac{x}{2}(\cos(\ln x) + \sin(\ln x)) + C}$$

$$(j) \int x^2 \ln x dx = \boxed{\frac{x^3}{3} \ln x - \frac{x^3}{9} + C}$$

$$(k) \int x^2 \cos^2 x dx = \boxed{\frac{x^3}{6} + \frac{x^2}{4} \sin(2x) + \frac{x}{4} \cos(2x) - \frac{1}{8} \sin(2x) + C}$$

5. Answers

$$(a) \int \frac{10 dx}{15 + 4x} = \boxed{\frac{5}{2} \ln |15 + 4x| + C}$$

$$(b) \int \frac{dx}{4 - 7x} = \boxed{-\frac{1}{7} \ln |4 - 7x| + C}$$

$$(c) \int \frac{dx}{(x + 15)^{20}} = \boxed{-\frac{1}{19(x + 15)^{19}} + C}$$

$$(d) \int \frac{dx}{(5 - 2x)^8} = \boxed{\frac{1}{14(5 - 2x)^7} + C}$$

$$(e) \int \frac{x}{2x^2 + 2x + 5} dx = \boxed{\frac{1}{4} \ln(2x^2 + 2x + 5) - \frac{1}{6} \arctan\left(\frac{2x + 1}{3}\right) + C}$$

$$(f) \int \frac{x}{x^2 - 7x + 20} dx = \boxed{\frac{1}{2} \ln(x^2 - 7x + 20) + \frac{7}{\sqrt{31}} \arctan\left(\frac{2x - 7}{\sqrt{31}}\right) + C}$$

$$(g) \int \frac{x + 1}{5x^2 + 2x + 1} dx = \boxed{\frac{1}{10} \ln(5x^2 + 2x + 1) + \frac{2}{5} \arctan\left(\frac{5x + 1}{2}\right) + C}$$

$$(h) \int \frac{2x+1}{x^2-4x+5} dx = \boxed{\ln(x^2-4x+5) + 5 \arctan(x-2) + C}$$

$$(i) \int \frac{dx}{2x^2-2x+3} = \boxed{\frac{1}{\sqrt{5}} \arctan\left(\frac{2x-1}{\sqrt{5}}\right) + C}$$

6. Answers (partial fractions)

$$(a) \int \frac{3x^3+5x^2-25x-1}{(x+2)(x-1)^2} dx = \boxed{3x + 5 \ln|x+2| + \frac{6}{x-1} + C}$$

$$(b) \int \frac{x^2-3x+17}{(x+5)(x^2+2x+4)} dx = \boxed{3 \ln|x+5| - \ln(x^2+2x+4) + \sqrt{3} \arctan\left(\frac{x+1}{\sqrt{3}}\right) + C}$$

$$(c) \int \frac{x^2+3}{(x+1)^2(x-1)} dx = \boxed{\frac{2}{x+1} + \ln|x-1| + C}$$

$$(d) \int \frac{x^3+2x+5}{x^2-1} dx = \boxed{\frac{x^2}{2} - \ln|x+1| + 4 \ln|x-1| + C}$$

$$(e) \int \frac{x^3+3x^2+5x+7}{x^2+2} dx = \boxed{\frac{x^2}{2} + 3x + \frac{3}{2} \ln(x^2+2) + \frac{1}{\sqrt{2}} \arctan\left(\frac{x}{\sqrt{2}}\right) + C}$$

$$(f) \int \frac{x^2+2x+6}{(x-1)(x-2)(x-4)} dx = \boxed{3 \ln|x-1| - 7 \ln|x-2| + 5 \ln|x-4| + C}$$

$$(g) \int \frac{2x^2+x+3}{(x+2)(x^2+x+1)} dx = \boxed{3 \ln|x+2| - \frac{1}{2} \ln(x^2+x+1) + \frac{1}{\sqrt{3}} \arctan\left(\frac{2x+1}{\sqrt{3}}\right) + C}$$

$$(h) \int \frac{dx}{x^5-x^2} = \boxed{\frac{1}{x} + \frac{1}{3} \ln|x-1| - \frac{1}{6} \ln(x^2+x+1) + \frac{\sqrt{3}}{3} \arctan\left(\frac{2x+1}{\sqrt{3}}\right) + C}$$

7. Answers (trigonometric)

$$(a) \int \frac{\cos x}{1+\cos x} dx = \boxed{x - \tan\left(\frac{x}{2}\right) + C}$$

$$(b) \int \frac{dx}{4 \sin x + 3 \cos x + 5} = \boxed{-\frac{1}{\tan(x/2) + 2} + C}$$

$$(c) \int \frac{dx}{5 \sin x + 3 \cos x + 3} = \boxed{\frac{1}{5} \ln \left| 3 + 5 \tan\left(\frac{x}{2}\right) \right| + C}$$

$$(d) \int \sin^5 x dx = \boxed{-\cos x + \frac{2}{3} \cos^3 x - \frac{1}{5} \cos^5 x + C}$$

$$(e) \int \sqrt[3]{\sin^2 x} \cos^3 x dx = \boxed{\frac{3}{5} \sin^{5/3} x - \frac{3}{11} \sin^{11/3} x + C}$$

$$(f) \int \sin^2 x \cos^2 x dx = \boxed{\frac{x}{8} - \frac{\sin(4x)}{32} + C}$$

$$(g) \int \sin^4(2x) dx = \boxed{\frac{3x}{8} - \frac{\sin(4x)}{8} + \frac{\sin(8x)}{64} + C}$$

$$(h) \int \sin(5x) \cos x \, dx = \boxed{-\frac{\cos(6x)}{12} - \frac{\cos(4x)}{8} + C}$$

$$(i) \int \cos(3x) \cos(2x) \, dx = \boxed{\frac{1}{2} \sin x + \frac{1}{10} \sin(5x) + C}$$

8. Answers (irrational)

$$(a) \int \frac{x+1}{\sqrt[3]{3x+1}} \, dx = \boxed{\frac{1}{15}(3x+1)^{5/3} + \frac{1}{3}(3x+1)^{2/3} + C}$$

$$(b) \int \frac{dx}{\sqrt[3]{(2x+1)^2} - \sqrt{2x+1}} = \boxed{\frac{3}{2}u^2 + 3u + 3 \ln |u-1| + C, \quad u = (2x+1)^{1/6}}$$

$$(c) \int \frac{\sqrt[6]{x}}{1 + \sqrt[3]{x}} \, dx = \boxed{\frac{6}{5}x^{5/6} - 2x^{1/2} + 6x^{1/6} - 6 \arctan(x^{1/6}) + C}$$

$$(d) \int \frac{7x+16}{\sqrt[3]{(7x+8)^2} + 2\sqrt[3]{7x+8}} \, dx = \boxed{\frac{3}{28}(7x+8)^{4/3} - \frac{2}{7}(7x+8) + \frac{6}{7}(7x+8)^{2/3} + C}$$